

52-WEEK COMPANION HANDBOOK

Professional reading guide, decision framework, and durable reference map for the complete zero-to-hero journey.

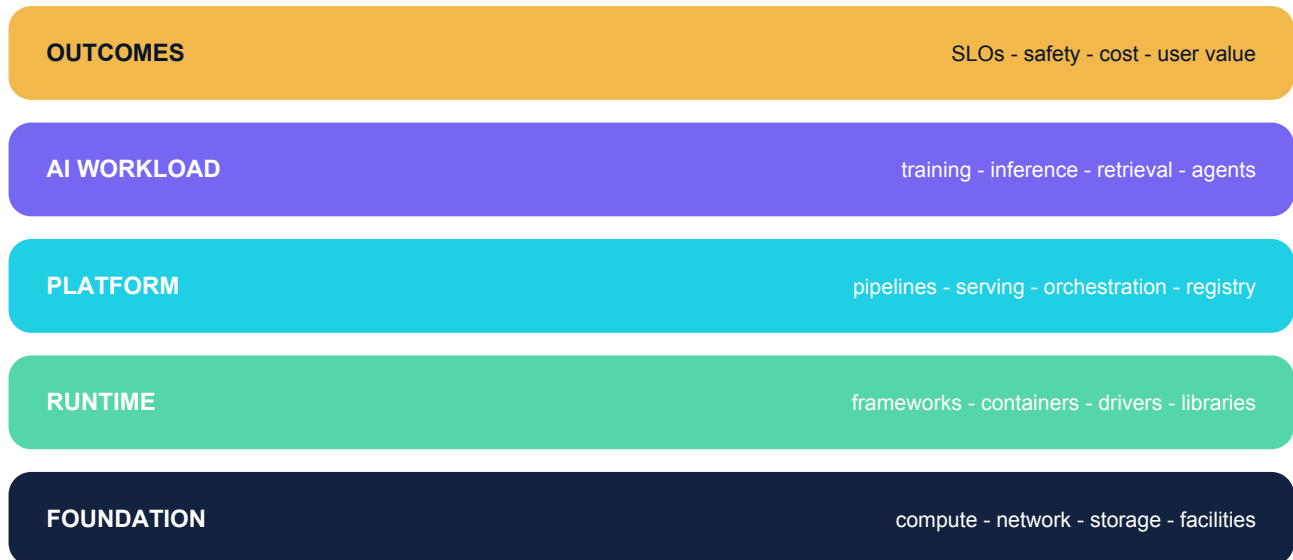


52 WEEKS	365 NUMBERED TOPICS	ZERO TO HERO
52 weekly guides	Professional resource edition	Concepts + practice + evidence

START WITH THE SYSTEM

AI infrastructure is the engineered path from data and code to a reliable, secure, cost-aware model outcome. The stack matters because every layer imposes constraints on the next: hardware shapes runtimes; runtimes shape scheduling; data paths shape accelerator utilization; serving contracts shape observability and reliability.

A strong infrastructure engineer does not begin with a product name. They begin with a workload, a boundary, a target, and evidence.



THE COURSE OPERATING LOOP

1	FRAME	Define users, workload shape, constraints, SLOs, security, and cost envelope.
2	DESIGN	Choose interfaces and failure domains before selecting implementations.
3	BUILD	Create a thin end-to-end slice with repeatable configuration.
4	MEASURE	Collect quality, latency, throughput, utilization, reliability, and cost evidence.
5	OPERATE	Alert, recover, patch, scale, audit, and improve from observed behavior.

LEARNING CONTRACT

Use these rules across every lecture and lab.

- Name the workload before the platform: training, batch inference, online inference, streaming, retrieval, simulation, or edge.
- Write measurable targets: quality, latency distribution, throughput, freshness, availability, recovery, privacy, and unit cost.
- Record versions and immutable identifiers for code, data, image, model, configuration, and infrastructure.
- Make failure visible with health checks, structured logs, metrics, traces, dashboards, alerts, and runbooks.
- Use least privilege, short-lived identity, protected secrets, encryption, provenance, and audit logs by default.
- Benchmark the end-to-end path on representative data and hardware; isolate bottlenecks before optimizing.
- Treat teardown, rollback, restore, and cost reporting as required parts of every build.

THE SIX NUMBERS TO CARRY

QUALITY	Task metric + segment risk	Does the model do the right thing?
LATENCY	p50 / p95 / p99	How slow is the tail?
THROUGHPUT	items, tokens, frames / second	How much useful work completes?
UTILIZATION	compute + memory + network	Where is capacity idle or saturated?
RELIABILITY	SLO + error budget + recovery	How often and how badly does it fail?
UNIT COST	cost / successful useful output	Is efficiency improving?

Introduction to AI Infrastructure

This week turns introduction to ai infrastructure into operational judgment. Focus on the relationship between workload requirements, system boundaries, failure behavior, and measurable evidence.

001	What Is AI Infrastructure? The Engine Behind AI
002	CPUs vs GPUs vs TPUs - Computing Power for AI
003	Training vs Inference - Two Faces of AI Workloads
004	AI Infrastructure Layers - Hardware, Software, Ops
005	Case Studies: Infrastructure Behind ChatGPT, DALL-E
006	Industry Landscape: Cloud Providers & AI Chips
007	Lab - Spin Up Your First AI VM

WEEKLY LENS

Build a mental model of the stack before choosing tools. Every layer should answer a workload requirement.

Production lens

Write the workload assumption, chosen control, measurable signal, failure mode, owner, and rollback or recovery step.

Working artifact

```
requirement -> option -> tradeoff ->
test -> decision record
```

Exit ticket

Sketch the system boundary. Name one constraint, one failure mode, one measurable signal, and one decision you can defend.

READ - BUILD - EXPLAIN

Read for the principle. Build the smallest artifact that exposes the principle. Explain the tradeoff using a diagram and measurements. If you cannot reproduce the result from a clean environment, the lab is not finished.

Linux Foundations for AI Engineers

This week turns linux foundations for ai engineers into operational judgment. Focus on the relationship between workload requirements, system boundaries, failure behavior, and measurable evidence.

008	Why Linux Dominates AI Infrastructure
009	Navigating the Linux Shell - Bash Basics
010	Filesystems, Directories, and Permissions
011	Package Managers - apt, yum, pip
012	Process Management and Monitoring
013	Scripting for Automation in AI Workflows
014	Lab - Set Up Ubuntu for AI Development

WEEKLY LENS

Use Linux as the observable, automatable substrate beneath drivers, runtimes, containers, and schedulers.

Production lens

Inspect identity, paths, permissions, processes, open ports, disk, memory, and logs before changing state.

Working artifact

```
id; pwd; df -h; free -h 2>/dev/null ||  
vm_stat; ps aux | head
```

Exit ticket

Sketch the system boundary. Name one constraint, one failure mode, one measurable signal, and one decision you can defend.

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Cloud Infrastructure Basics

This week turns cloud infrastructure basics into operational judgment. Focus on the relationship between workload requirements, system boundaries, failure behavior, and measurable evidence.

015	Introduction to Cloud for AI Workloads
016	Compute Instances - Choosing CPU vs GPU
017	Networking Basics - VPCs, Firewalls, Load Balancers
018	Cloud Storage - Object, Block, and File Systems
019	Hands-On with AWS EC2 for AI
020	Hands-On with Google Cloud GPU Instances
021	Lab - Compare Cost & Performance Across Clouds

WEEKLY LENS

Translate workload shape into compute, network, storage, availability, and cost decisions before provisioning.

Production lens

Verify quota and capacity, region, accelerator attachment, image and driver compatibility, network and storage path, and billing tags.

Working artifact

```
record region + instance + accelerator  
+ image + storage + hourly estimate
```

Exit ticket

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Containerization Foundations

This week turns containerization foundations into operational judgment. Focus on the relationship between workload requirements, system boundaries, failure behavior, and measurable evidence.

022	Why Containers Are Critical for AI
023	Docker Basics - Images and Containers
024	Building a Container for an AI App
025	Networking and Volumes in Docker
026	Docker Compose for Multi-Service AI Systems
027	Best Practices: Lightweight AI Containers
028	Lab - Containerize a PyTorch Model

WEEKLY LENS

Treat the image as a reproducible contract: pinned dependencies, a small attack surface, and a predictable entrypt.

Production lens

Verify pinned inputs, non-root execution, layer size, health behavior, and reproducibility from a clean build.

Working artifact

```
docker build -t ai-app:local . &&  
docker run --rm ai-app:local
```

Exit ticket

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Kubernetes Fundamentals

This week turns kubernetes fundamentals into operational judgment. Focus on the relationship between workload requirements, system boundaries, failure behavior, and measurable evidence.

029 What Is Kubernetes and Why AI Needs It

030 Pods, Nodes, and Clusters Explained

031 Deployments and Services for AI Apps

032 ConfigMaps, Secrets, and Volumes

033 Horizontal Pod Autoscaling for AI Inference

034 Helm Charts for Simplified AI Deployments

035 Lab - Deploy a Model on Minikube

WEEKLY LENS

Kubernetes reconciles declared state. Reliable AI platforms depend on correct requests, probes, storage, and rollout policy.

Production lens

Check requests and limits, probes, events, pending reasons, disruption policy, and rollback path.

Working artifact

```
kubect1 get pods -A -o wide && kubect1  
top pods -A
```

Exit ticket

Sketch the system boundary. Name one constraint, one failure mode, one measurable signal, and one decision you can defend.

READ - BUILD - EXPLAIN

Read for the principle. Build the smallest artifact that exposes the principle. Explain the tradeoff using a diagram and measurements. If you cannot reproduce the result from a clean environment, the lab is not finished.

Data Storage for AI

This week turns data storage for ai into operational judgment. Focus on the relationship between workload requirements, system boundaries, failure behavior, and measurable evidence.

036	Data Lakes vs Data Warehouses vs Feature Stores
037	Object Storage with AWS S3 and GCP GCS
038	Relational vs NoSQL Databases in AI
039	Streaming Data with Kafka & Pub/Sub
040	Scaling Storage for AI Training Datasets
041	Secure Data Access and Encryption
042	Lab - Build a Data Ingestion Pipeline

WEEKLY LENS

Separate durable source data, analytical transformations, and low-latency serving data while preserving lineage.

Production lens

Document schema, partitioning, retention, encryption, backup, restore time, lineage, and egress.

Working artifact

```
du -sh .; find . -type f | wc -l
```

Exit ticket

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GPU Hardware Deep Dive

This week turns gpu hardware deep dive into operational judgment. Focus on the relationship between workload requirements, system boundaries, failure behavior, and measurable evidence.

043	Anatomy of a GPU for AI
044	CUDA Basics and GPU Programming
045	GPU Memory Hierarchy - Optimizing Usage
046	Multi-GPU Scaling and Interconnects (NVLink)
047	Multi-Instance GPU (MIG) Configurations
048	Benchmarking AI Workloads on GPUs
049	Lab - Run a Model on GPU with CUDA

WEEKLY LENS

GPU performance is a balance of compute utilization, memory movement, communication, and thermal or power limits.

Production lens

Capture utilization, memory high-water mark, throughput, latency, power, and error state.

Working artifact

```
nvidia-smi --query-gpu=name,utilization.gpu,memory.used,memory.total
--format=csv
```

Exit ticket

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Distributed Training Basics

This week turns distributed training basics into operational judgment. Focus on the relationship between workload requirements, system boundaries, failure behavior, and measurable evidence.

050	Why Distributed Training Is Needed
051	Data Parallelism vs Model Parallelism
052	PyTorch Distributed Training
053	TensorFlow Multi-GPU Training
054	Horovod and AllReduce Explained
055	Fault Tolerance in Distributed Training
056	Lab - Train ResNet Across Multiple GPUs

WEEKLY LENS

Distributed speedup is earned only when useful compute outweighs communication, synchronization, and failure overhead.

Production lens

Report scaling efficiency, step time, collective time, stragglers, checkpoint duration, and recovery behavior.

Working artifact

```
efficiency = single_device_time /  
(devices * distributed_time)
```

Exit ticket

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Workflow Automation & Experiment Tracking

This week turns workflow automation & experiment tracking into operational judgment. Focus on the relationship between workload requirements, system boundaries, failure behavior, and measurable evidence.

057	Why Tracking ML Experiments Matters
058	Introduction to MLflow
059	Logging Metrics and Artifacts
060	Versioning Data, Models, and Parameters
061	Weights & Biases vs MLflow Comparison
062	Automating Training Pipelines
063	Lab - Track Experiments with MLflow

WEEKLY LENS

A result is trustworthy only when code, data, environment, parameters, metrics, and artifacts can be traced together.

Production lens

Confirm immutable identifiers, dataset reference, environment capture, metric definitions, artifact retention, and promotion evidence.

Working artifact

```
mlflow server --host 127.0.0.1 --port 5000
```

Exit ticket

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CI/CD for AI Models

This week turns ci/cd for ai models into operational judgment. Focus on the relationship between workload requirements, system boundaries, failure behavior, and measurable evidence.

064	What Is MLOps and CI/CD for AI?
065	GitHub Actions Basics for ML Projects
066	GitLab CI and Jenkins Pipelines
067	Testing Models Before Deployment
068	Automating Docker Builds for AI Apps
069	Canary Deployments and Rollbacks
070	Lab - CI/CD Pipeline for Model Deployment

WEEKLY LENS

CI/CD for models extends software delivery with data checks, model gates, reproducible images, and safe rollout evidence.

Production lens

Gate on unit, data, model, security, load, and smoke tests; record artifact digest and deployment evidence.

Working artifact

```
Build -> test -> scan -> sign ->
deploy -> verify -> promote/rollback
```

Exit ticket

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Advanced Kubernetes for AI

This week turns advanced kubernetes for ai into operational judgment. Focus on the relationship between workload requirements, system boundaries, failure behavior, and measurable evidence.

071	GPU Scheduling in Kubernetes
072	StatefulSets for Data-Heavy Workloads
073	Kubernetes Operators for ML
074	Helm Advanced Templates for AI Apps
075	Scaling AI Training with Kubeflow
076	Kubernetes Security Best Practices
077	Lab - Deploy MLflow on Kubernetes

WEEKLY LENS

Advanced Kubernetes work is policy work: scheduling, isolation, state, operators, templates, and platform guardrails.

Production lens

Check requests and limits, probes, events, pending reasons, disruption policy, and rollback path.

Working artifact

```
kubect1 get pods -A -o wide && kubect1  
top pods -A
```

Exit ticket

Sketch the system boundary. Name one constraint, one failure mode, one measurable signal, and one decision you can defend.

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Resource & Cost Optimization

This week turns resource & cost optimization into operational judgment. Focus on the relationship between workload requirements, system boundaries, failure behavior, and measurable evidence.

078	Why AI Infrastructure Costs Spike
079	Using Spot Instances for AI Training
080	Autoscaling AI Clusters in Cloud
081	Monitoring Resource Utilization
082	Storage Cost Optimization Strategies
083	Multi-Tenant Cost Allocation in Teams
084	Lab - Optimize Cloud AI Workload Costs

WEEKLY LENS

Optimize cost per useful outcome, not hourly price. Include utilization, retries, data movement, and engineer time.

Production lens

Attribute spend, normalize by workload output, inspect idle time, right-size, model interruptions, and set budgets.

Working artifact

```
unit cost = total run cost /
successful useful outputs
```

Exit ticket

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Networking for AI Systems

This week turns networking for ai systems into operational judgment. Focus on the relationship between workload requirements, system boundaries, failure behavior, and measurable evidence.

085	Fundamentals of Data Center Networking
086	Software Defined Networking (SDN) for AI
087	Infiniband and High-Speed Interconnects
088	Load Balancing for AI Inference
089	Network Bottlenecks in Distributed Training
090	Security in Networked AI Systems
091	Lab - Configure Load Balancer for AI API

WEEKLY LENS

AI networks carry control traffic, data loading, collective communication, and inference requests with different needs.

Production lens

Measure p50/p95/p99 latency, throughput, packet loss, retries, connection saturation, and cross-zone traffic.

Working artifact

```
curl -fsS -w '%{http_code}
%{time_total}\n' -o /dev/null
http://localhost:8000/health
```

Exit ticket

Sketch the system boundary. Name one constraint, one failure mode, one measurable signal, and one decision you can defend.

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Model Serving Basics

This week turns model serving basics into operational judgment. Focus on the relationship between workload requirements, system boundaries, failure behavior, and measurable evidence.

092	From Training to Serving - The Deployment Gap
093	REST vs gRPC for Model APIs
094	TensorFlow Serving for AI Models
095	TorchServe for PyTorch Models
096	Deploying Models with FastAPI
097	Scaling Model Serving with Kubernetes
098	Lab - Serve an Image Classifier with FastAPI

WEEKLY LENS

Serving turns a model artifact into a dependable product through a stable contract, packaging, scaling, and health signals.

Production lens

Test schema errors, readiness, concurrency, batching, timeouts, overload, model version, and rollback.

Working artifact

```
curl -fsS  
http://localhost:8000/health
```

Exit ticket

Sketch the system boundary. Name one constraint, one failure mode, one measurable signal, and one decision you can defend.

READ - BUILD - EXPLAIN

Read for the principle. Build the smallest artifact that exposes the principle. Explain the tradeoff using a diagram and measurements. If you cannot reproduce the result from a clean environment, the lab is not finished.

Advanced Model Serving

This week turns advanced model serving into operational judgment. Focus on the relationship between workload requirements, system boundaries, failure behavior, and measurable evidence.

099	NVIDIA TensorRT Optimization
100	Triton Inference Server Basics
101	Batch Inference vs Online Inference
102	Caching for Fast Inference
103	Multi-Model Serving Strategies
104	A/B Testing Model Endpoints
105	Lab - Deploy a Model on Triton Server

WEEKLY LENS

Advanced serving improves throughput and latency by aligning runtime, precision, batching, caching, and model placement.

Production lens

Test schema errors, readiness, concurrency, batching, timeouts, overload, model version, and rollback.

Working artifact

```
curl -fsS  
http://localhost:8000/health
```

Exit ticket

Sketch the system boundary. Name one constraint, one failure mode, one measurable signal, and one decision you can defend.

READ - BUILD - EXPLAIN

Read for the principle. Build the smallest artifact that exposes the principle. Explain the tradeoff using a diagram and measurements. If you cannot reproduce the result from a clean environment, the lab is not finished.

Observability in AI Infrastructure

This week turns observability in ai infrastructure into operational judgment. Focus on the relationship between workload requirements, system boundaries, failure behavior, and measurable evidence.

106	Why Monitoring AI Systems Matters
107	GPU Monitoring with DCGM
108	Metrics Collection with Prometheus
109	Visualization Dashboards with Grafana
110	Tracing AI Requests with OpenTelemetry
111	Building Alerts for AI System Failures
112	Lab - Monitor GPU Cluster with Prometheus

WEEKLY LENS

Observability connects resource health to model behavior and user impact through metrics, logs, traces, and actionable alerts.

Production lens

Define SLO signals, dashboards for decisions, trace context, runbook-linked alerts, and telemetry cost controls.

Working artifact

```
rate(requests_total[5m]);  
histogram_quantile(0.99, ...)
```

Exit ticket

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Model & Data Drift

This week turns model & data drift into operational judgment. Focus on the relationship between workload requirements, system boundaries, failure behavior, and measurable evidence.

- | | |
|-----|--|
| 113 | What Is Concept Drift vs Data Drift? |
| 114 | Why Drift Destroys AI Performance |
| 115 | Tools for Drift Detection (EvidentlyAI) |
| 116 | Real-Time Drift Monitoring Pipelines |
| 117 | Human-in-the-Loop Drift Evaluation |
| 118 | Mitigation Strategies - Retraining & Rebalancing |
| 119 | Lab - Build a Drift Detection Pipeline |

WEEKLY LENS

Drift monitoring compares present reality with a reference and turns statistically meaningful change into a response workflow.

Production lens

Track feature shift, prediction shift, delayed labels, segment impact, data quality, and retraining decisions.

Working artifact

```
reference window + current window +  
threshold + owner + action
```

Exit ticket

Sketch the system boundary. Name one constraint, one failure mode, one measurable signal, and one decision you can defend.

READ - BUILD - EXPLAIN

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AI Security & Compliance

This week turns ai security & compliance into operational judgment. Focus on the relationship between workload requirements, system boundaries, failure behavior, and measurable evidence.

120	Security Risks in AI Infrastructure
121	Identity and Access Management (IAM)
122	Secrets Management for AI Systems
123	Data Encryption at Rest and In Transit
124	Model Theft and Adversarial Attacks
125	Compliance Standards (GDPR, HIPAA, SOC2)
126	Lab - Secure a Model Endpoint with Authentication

WEEKLY LENS

**Secure AI infrastructure
with least privilege,
strong identity, secret
hygiene, encryption,
provenance, and
auditable controls.**

Production lens

Review principals, permissions, secret rotation, encryption keys, provenance, audit logs, threat assumptions, and incident steps.

Working artifact

who can do what to which resource
under which condition?

Exit ticket

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Reliability & High Availability

This week turns reliability & high availability into operational judgment. Focus on the relationship between workload requirements, system boundaries, failure behavior, and measurable evidence.

127	Why AI Systems Fail in Production
128	Fault Tolerance in AI Inference
129	Redundancy and Failover for AI APIs
130	Designing High-Availability AI Clusters
131	Auto-Healing Infrastructure with Kubernetes
132	Chaos Engineering for AI Systems
133	Lab - Build a HA AI Inference Cluster

WEEKLY LENS

Reliability comes from explicit failure assumptions, redundancy, bounded recovery, tested failover, and graceful degradation.

Production lens

Define SLI/SLO, RTO/RPO, dependency failure modes, retry budgets, health semantics, backup restore, and game days.

Working artifact

```
availability target -> error budget ->
controls -> test evidence
```

Exit ticket

Sketch the system boundary. Name one constraint, one failure mode, one measurable signal, and one decision you can defend.

READ - BUILD - EXPLAIN

Read for the principle. Build the smallest artifact that exposes the principle. Explain the tradeoff using a diagram and measurements. If you cannot reproduce the result from a clean environment, the lab is not finished.

Multi-Cloud AI Infrastructure

This week turns multi-cloud ai infrastructure into operational judgment. Focus on the relationship between workload requirements, system boundaries, failure behavior, and measurable evidence.

134	Why Enterprises Go Multi-Cloud
135	Kubernetes Federation Across Clouds
136	Data Replication Across Cloud Providers
137	Cost Arbitrage in Multi-Cloud AI
138	Disaster Recovery Across Regions
139	Hybrid On-Prem + Cloud AI Setups
140	Lab - Serve a Model Across AWS + GCP

WEEKLY LENS

Multi-cloud is justified by concrete resilience, governance, or market needs; portability always carries operational cost.

Production lens

Verify quota and capacity, region, accelerator attachment, image and driver compatibility, network and storage path, and billing tags.

Working artifact

```
record region + instance + accelerator  
+ image + storage + hourly estimate
```

Exit ticket

Sketch the system boundary. Name one constraint, one failure mode, one measurable signal, and one decision you can defend.

READ - BUILD - EXPLAIN

Read for the principle. Build the smallest artifact that exposes the principle. Explain the tradeoff using a diagram and measurements. If you cannot reproduce the result from a clean environment, the lab is not finished.

Edge AI Infrastructure Basics

This week turns edge ai infrastructure basics into operational judgment. Focus on the relationship between workload requirements, system boundaries, failure behavior, and measurable evidence.

141	What Is Edge AI and Why It Matters
142	Use Cases: Retail, Healthcare, Smart Cities
143	Introduction to NVIDIA Jetson Devices
144	Installing JetPack SDK on Jetson
145	Edge vs Cloud Tradeoffs for AI Deployment
146	ONNX Runtime for Edge Models
147	Lab - Deploy YOLOv5 on Jetson Nano

WEEKLY LENS

Edge systems trade elastic capacity for locality, privacy, disconnected operation, tight power budgets, and hardware diversity.

Production lens

Benchmark cold start, steady latency, peak memory, battery or power, thermals, accuracy, and offline behavior.

Working artifact

```
measure on target hardware - desktop
results are not a proxy
```

Exit ticket

Sketch the system boundary. Name one constraint, one failure mode, one measurable signal, and one decision you can defend.

READ - BUILD - EXPLAIN

Read for the principle. Build the smallest artifact that exposes the principle. Explain the tradeoff using a diagram and measurements. If you cannot reproduce the result from a clean environment, the lab is not finished.

Optimizing AI for Edge Devices

This week turns optimizing ai for edge devices into operational judgment. Focus on the relationship between workload requirements, system boundaries, failure behavior, and measurable evidence.

148	Quantization for Edge Efficiency
149	Pruning and Model Compression Basics
150	TensorRT for Edge Inference
151	Running Vision Models on Raspberry Pi
152	TinyML and Microcontrollers for AI
153	Benchmarking Models on Edge Hardware
154	Lab - Optimize a Model with TensorRT

WEEKLY LENS

Optimize edge models against a measured device budget: memory, latency, energy, thermals, and acceptable accuracy loss.

Production lens

Benchmark cold start, steady latency, peak memory, battery or power, thermals, accuracy, and offline behavior.

Working artifact

```
measure on target hardware - desktop  
results are not a proxy
```

Exit ticket

Sketch the system boundary. Name one constraint, one failure mode, one measurable signal, and one decision you can defend.

READ - BUILD - EXPLAIN

Read for the principle. Build the smallest artifact that exposes the principle. Explain the tradeoff using a diagram and measurements. If you cannot reproduce the result from a clean environment, the lab is not finished.

Mobile AI Infrastructure

This week turns mobile ai infrastructure into operational judgment. Focus on the relationship between workload requirements, system boundaries, failure behavior, and measurable evidence.

155	Why Mobile AI Is Booming
156	Core ML for iOS - Basics
157	TensorFlow Lite for Android - Basics
158	Deploying On-Device NLP Models
159	Power Management in Mobile AI
160	Privacy Considerations for On-Device AI
161	Lab - Build a Mobile AI App with TFLite

WEEKLY LENS

Mobile inference must respect platform runtimes, app lifecycle, privacy boundaries, battery, binary size, and device variance.

Production lens

Benchmark cold start, steady latency, peak memory, battery or power, thermals, accuracy, and offline behavior.

Working artifact

```
measure on target hardware - desktop  
results are not a proxy
```

Exit ticket

Sketch the system boundary. Name one constraint, one failure mode, one measurable signal, and one decision you can defend.

READ - BUILD - EXPLAIN

Read for the principle. Build the smallest artifact that exposes the principle. Explain the tradeoff using a diagram and measurements. If you cannot reproduce the result from a clean environment, the lab is not finished.

Data Pipelines for AI at Scale

This week turns data pipelines for ai at scale into operational judgment. Focus on the relationship between workload requirements, system boundaries, failure behavior, and measurable evidence.

162	Introduction to ETL and ELT Pipelines
163	Apache Airflow for AI Workflows
164	Streaming Data with Apache Kafka
165	Feature Stores for ML (Feast, Tecton)
166	Real-Time Data Preprocessing at Scale
167	Data Quality Monitoring for AI Systems
168	Lab - Build a Streaming Pipeline with Kafka

WEEKLY LENS

Data pipelines should be idempotent, observable, schema-aware, replayable, and aligned between training and serving.

Production lens

Track freshness, lag, duplicates, schema failures, dead-letter volume, replay time, and lineage.

Working artifact

```
Record: input -> transform -> output
      -> checkpoint -> alert
```

Exit ticket

Sketch the system boundary. Name one constraint, one failure mode, one measurable signal, and one decision you can defend.

READ - BUILD - EXPLAIN

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Generative AI Infrastructure – Foundations

This week turns generative ai infrastructure - foundations into operational judgment. Focus on the relationship between workload requirements, system boundaries, failure behavior, and measurable evidence.

169	Infrastructure Challenges of Large Language Models
170	Memory and Storage Needs of LLMs
171	Vector Databases - FAISS, Pinecone, Weaviate
172	RAG (Retrieval-Augmented Generation) Pipelines
173	Caching Strategies for LLMs
174	Serving LLMs in Production
175	Lab - Deploy a Simple RAG Pipeline

WEEKLY LENS

LLM systems are memory- and bandwidth-intensive; retrieval, caching, batching, and context policy often dominate economics.

Production lens

Write the workload assumption, chosen control, measurable signal, failure mode, owner, and rollback or recovery step.

Working artifact

```
requirement -> option -> tradeoff ->  
test -> decision record
```

Exit ticket

Sketch the system boundary. Name one constraint, one failure mode, one measurable signal, and one decision you can defend.

READ - BUILD - EXPLAIN

Read for the principle. Build the smallest artifact that exposes the principle. Explain the tradeoff using a diagram and measurements. If you cannot reproduce the result from a clean environment, the lab is not finished.

Generative AI Infrastructure – Advanced

This week turns generative ai infrastructure - advanced into operational judgment. Focus on the relationship between workload requirements, system boundaries, failure behavior, and measurable evidence.

176	DeepSpeed and ZeRO Optimization for LLMs
177	Megatron-LM for Large Model Training
178	Flash Attention and Memory Optimizations
179	Multi-Node Distributed Training for LLMs
180	Fine-Tuning vs Parameter-Efficient Tuning
181	Cost Challenges of Generative AI Training
182	Lab - Fine-Tune a Small LLM with PEFT

WEEKLY LENS

Large-model optimization shards state and reduces memory movement while keeping training numerically stable and recoverable.

Production lens

Write the workload assumption, chosen control, measurable signal, failure mode, owner, and rollback or recovery step.

Working artifact

```
requirement -> option -> tradeoff ->
test -> decision record
```

Exit ticket

Sketch the system boundary. Name one constraint, one failure mode, one measurable signal, and one decision you can defend.

READ - BUILD - EXPLAIN

Read for the principle. Build the smallest artifact that exposes the principle. Explain the tradeoff using a diagram and measurements. If you cannot reproduce the result from a clean environment, the lab is not finished.

Infrastructure for Computer Vision at Scale

This week turns infrastructure for computer vision at scale into operational judgment. Focus on the relationship between workload requirements, system boundaries, failure behavior, and measurable evidence.

183	Image and Video Data Challenges in AI Infra
184	Training Pipelines for CV Models
185	EfficientNet and Model Scaling Tradeoffs
186	Serving Real-Time Video Inference
187	DeepStream SDK for Video Analytics
188	Edge-to-Cloud Video Processing
189	Lab - Deploy Real-Time Object Detection

WEEKLY LENS

Computer-vision infrastructure must move and decode high-volume media efficiently before the accelerator can stay busy.

Production lens

Measure frames per second, dropped frames, decode time, queue depth, transfer time, latency, and task quality.

Working artifact

```
camera/data -> decode -> preprocess ->
infer -> postprocess -> sink
```

Exit ticket

Sketch the system boundary. Name one constraint, one failure mode, one measurable signal, and one decision you can defend.

READ - BUILD - EXPLAIN

Read for the principle. Build the smallest artifact that exposes the principle. Explain the tradeoff using a diagram and measurements. If you cannot reproduce the result from a clean environment, the lab is not finished.

Infrastructure for NLP at Scale

This week turns infrastructure for nlp at scale into operational judgment. Focus on the relationship between workload requirements, system boundaries, failure behavior, and measurable evidence.

190	NLP Workloads: Tokenization, Embeddings, Transformers
191	Training Large Transformer Models
192	Infrastructure for BERT and GPT
193	Efficient Serving of Embedding Models
194	Latency Reduction in NLP Inference
195	Deploying Multilingual Models at Scale
196	Lab - Deploy a BERT Model with FastAPI

WEEKLY LENS

NLP systems balance sequence length, tokenizer cost, embedding throughput, accelerator memory, and latency targets.

Production lens

Measure retrieval quality, token volume, time to first token, output rate, tail latency, cache hit rate, and groundedness.

Working artifact

```
request -> retrieve -> rerank ->
assemble -> generate -> evaluate
```

Exit ticket

Sketch the system boundary. Name one constraint, one failure mode, one measurable signal, and one decision you can defend.

READ - BUILD - EXPLAIN

Read for the principle. Build the smallest artifact that exposes the principle. Explain the tradeoff using a diagram and measurements. If you cannot reproduce the result from a clean environment, the lab is not finished.

Infrastructure for Multimodal AI

This week turns infrastructure for multimodal ai into operational judgment. Focus on the relationship between workload requirements, system boundaries, failure behavior, and measurable evidence.

197	What Is Multimodal AI?
198	Handling Text + Image Pipelines
199	Training and Serving CLIP Models
200	Infrastructure for Speech + Text Models
201	Deploying Video + Text Search Systems
202	Challenges of Multimodal Model Serving
203	Lab - Deploy CLIP for Image Search

WEEKLY LENS

Multimodal systems need explicit contracts between encoders, modalities, indexes, and asynchronous processing stages.

Production lens

Measure frames per second, dropped frames, decode time, queue depth, transfer time, latency, and task quality.

Working artifact

```
camera/data -> decode -> preprocess ->
infer -> postprocess -> sink
```

Exit ticket

Sketch the system boundary. Name one constraint, one failure mode, one measurable signal, and one decision you can defend.

READ - BUILD - EXPLAIN

Read for the principle. Build the smallest artifact that exposes the principle. Explain the tradeoff using a diagram and measurements. If you cannot reproduce the result from a clean environment, the lab is not finished.

Infrastructure for Reinforcement Learning

This week turns infrastructure for reinforcement learning into operational judgment. Focus on the relationship between workload requirements, system boundaries, failure behavior, and measurable evidence.

204	Basics of Reinforcement Learning Workloads
205	Simulation Environments for RL (Gym, Isaac)
206	Distributed RL Training Infrastructures
207	Real-Time Serving of RL Agents
208	Scaling RL for Robotics
209	Infrastructure for Online Learning Agents
210	Lab - Train RL Agent with Ray RLlib

WEEKLY LENS

Reinforcement learning infrastructure couples simulation throughput, experience collection, learner updates, and policy serving.

Production lens

Track environment steps/sec, queue age, learner utilization, reward stability, policy lag, and simulation fidelity.

Working artifact

```
actors -> experience -> replay ->
learner -> policy distribution
```

Exit ticket

Sketch the system boundary. Name one constraint, one failure mode, one measurable signal, and one decision you can defend.

READ - BUILD - EXPLAIN

Read for the principle. Build the smallest artifact that exposes the principle. Explain the tradeoff using a diagram and measurements. If you cannot reproduce the result from a clean environment, the lab is not finished.

Large-Scale Training – Basics

This week turns large-scale training - basics into operational judgment. Focus on the relationship between workload requirements, system boundaries, failure behavior, and measurable evidence.

211	What Is Large-Scale Training?
212	Data Parallelism vs Model Parallelism Revisited
213	Pipeline Parallelism in Transformers
214	Distributed Optimizers and Gradient Sync
215	Infrastructure Bottlenecks in Training LLMs
216	Fault Tolerance in Multi-Node Training
217	Lab - Train Transformer Across Multiple Nodes

WEEKLY LENS

Choose parallelism from the model, batch, topology, and memory envelope; measure scaling efficiency at every step.

Production lens

Write the workload assumption, chosen control, measurable signal, failure mode, owner, and rollback or recovery step.

Working artifact

```
requirement -> option -> tradeoff ->
test -> decision record
```

Exit ticket

Sketch the system boundary. Name one constraint, one failure mode, one measurable signal, and one decision you can defend.

READ - BUILD - EXPLAIN

Read for the principle. Build the smallest artifact that exposes the principle. Explain the tradeoff using a diagram and measurements. If you cannot reproduce the result from a clean environment, the lab is not finished.

Large-Scale Training – Advanced

This week turns large-scale training - advanced into operational judgment. Focus on the relationship between workload requirements, system boundaries, failure behavior, and measurable evidence.

218	DeepSpeed ZeRO-2/3 Optimizations
219	Fully Sharded Data Parallel (FSDP)
220	Flash Attention in Large Models
221	Checkpointing Strategies for Multi-Node Systems
222	Elastic Training with Kubernetes
223	Scaling Beyond 1,000 GPUs
224	Lab - Train with DeepSpeed ZeRO-3

WEEKLY LENS

At extreme scale, checkpoint design, elastic recovery, topology-aware collectives, and scheduler behavior become first-class.

Production lens

Write the workload assumption, chosen control, measurable signal, failure mode, owner, and rollback or recovery step.

Working artifact

```
requirement -> option -> tradeoff ->
test -> decision record
```

Exit ticket

Sketch the system boundary. Name one constraint, one failure mode, one measurable signal, and one decision you can defend.

READ - BUILD - EXPLAIN

Read for the principle. Build the smallest artifact that exposes the principle. Explain the tradeoff using a diagram and measurements. If you cannot reproduce the result from a clean environment, the lab is not finished.

Enterprise MLOps – Foundations

This week turns enterprise mlops - foundations into operational judgment. Focus on the relationship between workload requirements, system boundaries, failure behavior, and measurable evidence.

225	What Is Enterprise MLOps?
226	Introduction to Kubeflow
227	Introduction to MLflow at Scale
228	SageMaker Pipelines - Basics
229	GCP Vertex AI Pipelines
230	Azure ML Pipelines
231	Lab - Build a Pipeline in Kubeflow

WEEKLY LENS

Enterprise MLOps standardizes the paved road from data to deployment while preserving choice behind stable interfaces.

Production lens

Write the workload assumption, chosen control, measurable signal, failure mode, owner, and rollback or recovery step.

Working artifact

```
requirement -> option -> tradeoff ->
test -> decision record
```

Exit ticket

Sketch the system boundary. Name one constraint, one failure mode, one measurable signal, and one decision you can defend.

READ - BUILD - EXPLAIN

Read for the principle. Build the smallest artifact that exposes the principle. Explain the tradeoff using a diagram and measurements. If you cannot reproduce the result from a clean environment, the lab is not finished.

Enterprise MLOps – Advanced

This week turns enterprise mlops - advanced into operational judgment. Focus on the relationship between workload requirements, system boundaries, failure behavior, and measurable evidence.

232	Model Registry in Enterprise MLOps
233	Continuous Training (CT) Pipelines
234	Automating Drift Retraining with Kubeflow
235	Feature Store Integration in Pipelines
236	Governance in Enterprise MLOps
237	Audit Trails and Compliance Logging
238	Lab - Automate Drift Retraining with Kubeflow

WEEKLY LENS

Governed automation needs registries, approvals, lineage, policy checks, auditability, and repeatable retraining triggers.

Production lens

Write the workload assumption, chosen control, measurable signal, failure mode, owner, and rollback or recovery step.

Working artifact

```
requirement -> option -> tradeoff ->
test -> decision record
```

Exit ticket

Sketch the system boundary. Name one constraint, one failure mode, one measurable signal, and one decision you can defend.

READ - BUILD - EXPLAIN

Read for the principle. Build the smallest artifact that exposes the principle. Explain the tradeoff using a diagram and measurements. If you cannot reproduce the result from a clean environment, the lab is not finished.

Optimization Techniques – Foundations

This week turns optimization techniques - foundations into operational judgment. Focus on the relationship between workload requirements, system boundaries, failure behavior, and measurable evidence.

239	What Is Model Optimization for Infra Efficiency?
240	Quantization Basics
241	Pruning Basics
242	Distillation Basics
243	Structured vs Unstructured Sparsity
244	Benchmarking Optimized Models
245	Lab - Quantize a Vision Model

WEEKLY LENS

Optimization is an empirical trade: reduce precision, parameters, or computation only after defining an accuracy budget.

Production lens

Write the workload assumption, chosen control, measurable signal, failure mode, owner, and rollback or recovery step.

Working artifact

```
requirement -> option -> tradeoff ->
test -> decision record
```

Exit ticket

Sketch the system boundary. Name one constraint, one failure mode, one measurable signal, and one decision you can defend.

READ - BUILD - EXPLAIN

Read for the principle. Build the smallest artifact that exposes the principle. Explain the tradeoff using a diagram and measurements. If you cannot reproduce the result from a clean environment, the lab is not finished.

Optimization Techniques – Advanced

This week turns optimization techniques - advanced into operational judgment. Focus on the relationship between workload requirements, system boundaries, failure behavior, and measurable evidence.

246	Mixed Precision Training with AMP
247	Quantization-Aware Training (QAT)
248	Advanced Distillation Techniques
249	Sparse Training and Hardware Impacts
250	Compiler Optimizations (XLA, TorchDynamo)
251	Infra Tradeoffs: Accuracy vs Efficiency
252	Lab - Train with Mixed Precision

WEEKLY LENS

Advanced efficiency combines numerical formats, compiler graphs, sparsity, and hardware support; benchmark the full path.

Production lens

Write the workload assumption, chosen control, measurable signal, failure mode, owner, and rollback or recovery step.

Working artifact

```
requirement -> option -> tradeoff ->
test -> decision record
```

Exit ticket

Sketch the system boundary. Name one constraint, one failure mode, one measurable signal, and one decision you can defend.

READ - BUILD - EXPLAIN

Read for the principle. Build the smallest artifact that exposes the principle. Explain the tradeoff using a diagram and measurements. If you cannot reproduce the result from a clean environment, the lab is not finished.

Federated Learning Infrastructure

This week turns federated learning infrastructure into operational judgment. Focus on the relationship between workload requirements, system boundaries, failure behavior, and measurable evidence.

253	What Is Federated Learning?
254	Privacy-Preserving AI at Scale
255	Federated Learning with TensorFlow Federated
256	Secure Aggregation Protocols
257	Federated Data Challenges
258	Edge Deployment of Federated Models
259	Lab - Train a Federated Model with TFF

WEEKLY LENS

Federated learning moves computation toward data and introduces device orchestration, aggregation, privacy, and skew challenges.

Production lens

Record trust assumptions, privacy budget or protocol, client skew, dropout behavior, aggregation integrity, and utility loss.

Working artifact

threat -> control -> residual risk ->
utility cost -> evidence

Exit ticket

Sketch the system boundary. Name one constraint, one failure mode, one measurable signal, and one decision you can defend.

READ - BUILD - EXPLAIN

Read for the principle. Build the smallest artifact that exposes the principle. Explain the tradeoff using a diagram and measurements. If you cannot reproduce the result from a clean environment, the lab is not finished.

Privacy-Preserving AI

This week turns privacy-preserving ai into operational judgment. Focus on the relationship between workload requirements, system boundaries, failure behavior, and measurable evidence.

260	Why Privacy Matters in AI Infra
261	Homomorphic Encryption Basics
262	Differential Privacy for AI Models
263	Secure Multi-Party Computation (MPC)
264	Tradeoffs in Privacy-Preserving AI
265	Industry Applications of Privacy-Preserving AI
266	Lab - Apply Differential Privacy in Training

WEEKLY LENS

Privacy techniques change utility, performance, and threat models; choose controls from explicit exposure assumptions.

Production lens

Review principals, permissions, secret rotation, encryption keys, provenance, audit logs, threat assumptions, and incident steps.

Working artifact

who can do what to which resource
under which condition?

Exit ticket

Sketch the system boundary. Name one constraint, one failure mode, one measurable signal, and one decision you can defend.

READ - BUILD - EXPLAIN

Read for the principle. Build the smallest artifact that exposes the principle. Explain the tradeoff using a diagram and measurements. If you cannot reproduce the result from a clean environment, the lab is not finished.

AI Infrastructure Security – Advanced

This week turns ai infrastructure security - advanced into operational judgment. Focus on the relationship between workload requirements, system boundaries, failure behavior, and measurable evidence.

267	Attacks on AI Infrastructure
268	Model Poisoning Attacks
269	Data Poisoning Attacks
270	Membership Inference Attacks
271	Adversarial Examples in Deployment
272	Mitigation Strategies for Infra Security
273	Lab - Defend Against Adversarial Attacks

WEEKLY LENS

Defend the AI supply chain and runtime against poisoned inputs, stolen artifacts, inference leakage, and adversarial behavior.

Production lens

Review principals, permissions, secret rotation, encryption keys, provenance, audit logs, threat assumptions, and incident steps.

Working artifact

who can do what to which resource
under which condition?

Exit ticket

Sketch the system boundary. Name one constraint, one failure mode, one measurable signal, and one decision you can defend.

READ - BUILD - EXPLAIN

Read for the principle. Build the smallest artifact that exposes the principle. Explain the tradeoff using a diagram and measurements. If you cannot reproduce the result from a clean environment, the lab is not finished.

Multi-Tenant AI Infrastructure

This week turns multi-tenant ai infrastructure into operational judgment. Focus on the relationship between workload requirements, system boundaries, failure behavior, and measurable evidence.

274	What Is Multi-Tenancy?
275	Resource Sharing Across Teams
276	Cost Allocation for Multi-Tenant Infra
277	Role-Based Access Control (RBAC)
278	Isolation Strategies in AI Systems
279	Monitoring Multi-Tenant Environments
280	Lab - Configure Multi-Tenant Cluster

WEEKLY LENS

Multi-tenancy requires measurable fairness, hard isolation where needed, delegated access, quotas, and per-tenant telemetry.

Production lens

Write the workload assumption, chosen control, measurable signal, failure mode, owner, and rollback or recovery step.

Working artifact

```
requirement -> option -> tradeoff ->
test -> decision record
```

Exit ticket

Sketch the system boundary. Name one constraint, one failure mode, one measurable signal, and one decision you can defend.

READ - BUILD - EXPLAIN

Read for the principle. Build the smallest artifact that exposes the principle. Explain the tradeoff using a diagram and measurements. If you cannot reproduce the result from a clean environment, the lab is not finished.

AI Infrastructure for Startups

This week turns ai infrastructure for startups into operational judgment. Focus on the relationship between workload requirements, system boundaries, failure behavior, and measurable evidence.

281	Challenges Startups Face in AI Infra
282	Lean GPU Cloud Solutions
283	Open-Source MLOps Tools for Startups
284	Budget Optimization for Small Teams
285	Scaling from MVP to Production AI
286	Vendor Lock-In Risks for Startups
287	Lab - Deploy AI Infra on Low Budget

WEEKLY LENS

Startup infrastructure should buy learning speed first, add reliability deliberately, and avoid commitments that block change.

Production lens

Write the workload assumption, chosen control, measurable signal, failure mode, owner, and rollback or recovery step.

Working artifact

```
requirement -> option -> tradeoff ->
test -> decision record
```

Exit ticket

Sketch the system boundary. Name one constraint, one failure mode, one measurable signal, and one decision you can defend.

READ - BUILD - EXPLAIN

Read for the principle. Build the smallest artifact that exposes the principle. Explain the tradeoff using a diagram and measurements. If you cannot reproduce the result from a clean environment, the lab is not finished.

AI Infrastructure for Enterprises

This week turns ai infrastructure for enterprises into operational judgment. Focus on the relationship between workload requirements, system boundaries, failure behavior, and measurable evidence.

288	Enterprise-Scale Infra Needs
289	Vendor Selection for AI Infrastructure
290	Hybrid On-Prem + Cloud AI Strategy
291	Compliance and Regulatory Burdens
292	Integration with Enterprise IT Systems
293	Scaling Teams for AI Infra Management
294	Lab - Design Enterprise AI Infra Plan

WEEKLY LENS

Enterprise platforms optimize for governance, integration, predictable operations, and organizational scale as well as compute.

Production lens

Write the workload assumption, chosen control, measurable signal, failure mode, owner, and rollback or recovery step.

Working artifact

```
requirement -> option -> tradeoff ->
test -> decision record
```

Exit ticket

Sketch the system boundary. Name one constraint, one failure mode, one measurable signal, and one decision you can defend.

READ - BUILD - EXPLAIN

Read for the principle. Build the smallest artifact that exposes the principle. Explain the tradeoff using a diagram and measurements. If you cannot reproduce the result from a clean environment, the lab is not finished.

Infrastructure for Real-Time AI

This week turns infrastructure for real-time ai into operational judgment. Focus on the relationship between workload requirements, system boundaries, failure behavior, and measurable evidence.

295	Real-Time AI Use Cases (Ads, Fraud, Personalization)
296	Latency Challenges in Real-Time AI
297	Streaming Infrastructure for Real-Time AI
298	Deploying Low-Latency APIs
299	Scaling Real-Time Recommendation Systems
300	Cost Challenges in Real-Time AI
301	Lab - Build Real-Time Fraud Detection Pipeline

WEEKLY LENS

Real-time AI begins with a latency budget that assigns time and failure behavior to every hop in the serving path.

Production lens

Write the workload assumption, chosen control, measurable signal, failure mode, owner, and rollback or recovery step.

Working artifact

```
requirement -> option -> tradeoff ->
test -> decision record
```

Exit ticket

Sketch the system boundary. Name one constraint, one failure mode, one measurable signal, and one decision you can defend.

READ - BUILD - EXPLAIN

Read for the principle. Build the smallest artifact that exposes the principle. Explain the tradeoff using a diagram and measurements. If you cannot reproduce the result from a clean environment, the lab is not finished.

Infrastructure for Autonomous Systems

This week turns infrastructure for autonomous systems into operational judgment. Focus on the relationship between workload requirements, system boundaries, failure behavior, and measurable evidence.

302	AI Infra in Self-Driving Cars
303	Robotics AI Infrastructure Basics
304	Sensor Fusion Data Pipelines
305	Real-Time AI in Safety-Critical Systems
306	Simulation Infrastructure for Autonomous Vehicles
307	Edge Deployment in Robotics
308	Lab - Deploy AI Agent for Robotics Simulation

WEEKLY LENS

Autonomous systems combine safety constraints, sensor time alignment, simulation, edge deployment, and rigorous validation.

Production lens

Track environment steps/sec, queue age, learner utilization, reward stability, policy lag, and simulation fidelity.

Working artifact

```
actors -> experience -> replay ->
learner -> policy distribution
```

Exit ticket

Sketch the system boundary. Name one constraint, one failure mode, one measurable signal, and one decision you can defend.

READ - BUILD - EXPLAIN

Read for the principle. Build the smallest artifact that exposes the principle. Explain the tradeoff using a diagram and measurements. If you cannot reproduce the result from a clean environment, the lab is not finished.

AI Infrastructure – Case Studies

This week turns ai infrastructure - case studies into operational judgment. Focus on the relationship between workload requirements, system boundaries, failure behavior, and measurable evidence.

309	Case Study: OpenAI Infra for GPT
310	Case Study: Google Infra for DeepMind
311	Case Study: Meta Infra for LLaMA
312	Case Study: Tesla AI Infra for FSD
313	Case Study: Netflix AI Infra for Recommendations
314	Case Study: Healthcare AI Infra at Scale
315	Lab - Analyze Case Study Infra Architecture

WEEKLY LENS

Case studies are evidence, not templates: identify workload, constraints, tradeoffs, and what would differ in your context.

Production lens

Write the workload assumption, chosen control, measurable signal, failure mode, owner, and rollback or recovery step.

Working artifact

```
requirement -> option -> tradeoff ->
test -> decision record
```

Exit ticket

Sketch the system boundary. Name one constraint, one failure mode, one measurable signal, and one decision you can defend.

READ - BUILD - EXPLAIN

Read for the principle. Build the smallest artifact that exposes the principle. Explain the tradeoff using a diagram and measurements. If you cannot reproduce the result from a clean environment, the lab is not finished.

Future of AI Infrastructure

This week turns future of ai infrastructure into operational judgment. Focus on the relationship between workload requirements, system boundaries, failure behavior, and measurable evidence.

316	Trends in AI Chips (GPUs, TPUs, NPUs)
317	Cloud Evolution for AI
318	AI + Quantum Computing Infrastructure
319	Software Trends in AI Infra (Rust, Mojo)
320	Green AI and Sustainable Infrastructure
321	Global AI Regulation Impact on Infra
322	Lab - Design Future-Proof AI Infra

WEEKLY LENS

Future-proof architecture favors clear interfaces, measurable efficiency, replaceable components, and regulatory readiness.

Production lens

Write the workload assumption, chosen control, measurable signal, failure mode, owner, and rollback or recovery step.

Working artifact

```
requirement -> option -> tradeoff ->
test -> decision record
```

Exit ticket

Sketch the system boundary. Name one constraint, one failure mode, one measurable signal, and one decision you can defend.

READ - BUILD - EXPLAIN

Read for the principle. Build the smallest artifact that exposes the principle. Explain the tradeoff using a diagram and measurements. If you cannot reproduce the result from a clean environment, the lab is not finished.

Pre-Capstone Prep – Review

This week turns pre-capstone prep - review into operational judgment. Focus on the relationship between workload requirements, system boundaries, failure behavior, and measurable evidence.

323	Review of Hardware Concepts
324	Review of Cloud Infrastructure
325	Review of Containerization and Kubernetes
326	Review of MLOps Pipelines
327	Review of Monitoring and Security
328	Review of Cost Optimization Strategies
329	Lab - Mini-Project Review Sprint

WEEKLY LENS

Use review to connect layers: workload -> platform -> delivery -> observability -> security -> economics.

Production lens

Show requirements, diagram, reproducible demo, SLO results, cost model, failure test, security posture, and limitations.

Working artifact

```
claim -> metric -> method -> result ->
decision
```

Exit ticket

Sketch the system boundary. Name one constraint, one failure mode, one measurable signal, and one decision you can defend.

READ - BUILD - EXPLAIN

Read for the principle. Build the smallest artifact that exposes the principle. Explain the tradeoff using a diagram and measurements. If you cannot reproduce the result from a clean environment, the lab is not finished.

Capstone – Problem Definition

This week turns capstone - problem definition into operational judgment. Focus on the relationship between workload requirements, system boundaries, failure behavior, and measurable evidence.

330	Choosing a Capstone Domain (NLP, Vision, Generative AI)
331	Defining Success Metrics for Infra Project
332	Designing Initial Infrastructure Blueprint
333	Estimating Hardware and Cloud Costs
334	Selecting Tools and Frameworks for Build
335	Identifying Risks and Mitigation Plans
336	Lab - Capstone Project Proposal

WEEKLY LENS

A strong capstone starts with a bounded user problem, explicit SLOs, testable hypotheses, and a cost envelope.

Production lens

Show requirements, diagram, reproducible demo, SLO results, cost model, failure test, security posture, and limitations.

Working artifact

```
claim -> metric -> method -> result ->
decision
```

Exit ticket

Sketch the system boundary. Name one constraint, one failure mode, one measurable signal, and one decision you can defend.

READ - BUILD - EXPLAIN

Read for the principle. Build the smallest artifact that exposes the principle. Explain the tradeoff using a diagram and measurements. If you cannot reproduce the result from a clean environment, the lab is not finished.

Capstone – Implementation Phase I

This week turns capstone - implementation phase i into operational judgment. Focus on the relationship between workload requirements, system boundaries, failure behavior, and measurable evidence.

337	Setting Up Base Cloud/GPU Environment
338	Containerizing Models for Deployment
339	Setting Up Orchestration with Kubernetes
340	Configuring Storage and Data Pipelines
341	Implementing CI/CD for Model Deployment
342	Building Initial Monitoring Dashboards
343	Lab - Deploy First Working Infra Prototype

WEEKLY LENS

Build the thinnest end-to-end slice first so infrastructure, deployment, data, and telemetry fail together early.

Production lens

Show requirements, diagram, reproducible demo, SLO results, cost model, failure test, security posture, and limitations.

Working artifact

```
claim -> metric -> method -> result ->
decision
```

Exit ticket

Sketch the system boundary. Name one constraint, one failure mode, one measurable signal, and one decision you can defend.

READ - BUILD - EXPLAIN

Read for the principle. Build the smallest artifact that exposes the principle. Explain the tradeoff using a diagram and measurements. If you cannot reproduce the result from a clean environment, the lab is not finished.

Capstone – Implementation Phase II

This week turns capstone - implementation phase ii into operational judgment. Focus on the relationship between workload requirements, system boundaries, failure behavior, and measurable evidence.

344	Scaling Training Across Multi-GPU Nodes
345	Implementing Drift Detection Pipeline
346	Securing Model Endpoints with IAM
347	Cost Monitoring and Optimization Setup
348	Configuring High Availability Clusters
349	Stress Testing the AI System
350	Lab - Deliver Scalable Infra Deployment

WEEKLY LENS

Scale and harden only after baselining: load test, add drift and access controls, inspect cost, then prove recovery.

Production lens

Show requirements, diagram, reproducible demo, SLO results, cost model, failure test, security posture, and limitations.

Working artifact

```
claim -> metric -> method -> result ->
decision
```

Exit ticket

Sketch the system boundary. Name one constraint, one failure mode, one measurable signal, and one decision you can defend.

READ - BUILD - EXPLAIN

Read for the principle. Build the smallest artifact that exposes the principle. Explain the tradeoff using a diagram and measurements. If you cannot reproduce the result from a clean environment, the lab is not finished.

Capstone – Finalization

This week turns capstone - finalization into operational judgment. Focus on the relationship between workload requirements, system boundaries, failure behavior, and measurable evidence.

351	Conducting End-to-End Testing
352	Adding Redundancy and Failover Mechanisms
353	Polishing Monitoring and Alerting Dashboards
354	Generating Documentation for AI Infra
355	Creating Demo API Endpoints
356	Preparing Final Capstone Report
357	Lab - Capstone Infra Walkthrough

WEEKLY LENS

Finalization converts a prototype into an explainable system with tests, runbooks, redundancy, documentation, and a demo path.

Production lens

Show requirements, diagram, reproducible demo, SLO results, cost model, failure test, security posture, and limitations.

Working artifact

```
claim -> metric -> method -> result ->
decision
```

Exit ticket

Sketch the system boundary. Name one constraint, one failure mode, one measurable signal, and one decision you can defend.

READ - BUILD - EXPLAIN

Read for the principle. Build the smallest artifact that exposes the principle. Explain the tradeoff using a diagram and measurements. If you cannot reproduce the result from a clean environment, the lab is not finished.

Capstone – Presentation & Graduation

This week turns capstone - presentation & graduation into operational judgment. Focus on the relationship between workload requirements, system boundaries, failure behavior, and measurable evidence.

358	Preparing Final Demo Environment
359	Recording Metrics and Performance Benchmarks
360	Building Visuals of Infra Architecture
361	Delivering Stakeholder Presentation
362	Peer Review and Feedback on Projects
363	Reflection on Zero-to-Hero Journey
364	Future Learning Paths in AI Infrastructure
365	Graduation & Certification Showcase

WEEKLY LENS

Present decisions through evidence: requirements, architecture, benchmark results, cost, risks, and the next engineering step.

Production lens

Show requirements, diagram, reproducible demo, SLO results, cost model, failure test, security posture, and limitations.

Working artifact

```
claim -> metric -> method -> result ->
decision
```

Exit ticket

Sketch the system boundary. Name one constraint, one failure mode, one measurable signal, and one decision you can defend.

READ - BUILD - EXPLAIN

Read for the principle. Build the smallest artifact that exposes the principle. Explain the tradeoff using a diagram and measurements. If you cannot reproduce the result from a clean environment, the lab is not finished.

MAINTAINED REFERENCE MAP

Use first-party documentation for setup details and version-sensitive behavior. Record the date and version consulted in each lab report.

Docker documentation	https://docs.docker.com/get-started/
Kubernetes concepts	https://kubernetes.io/docs/concepts/
PyTorch documentation	https://docs.pytorch.org/docs/stable/
PyTorch distributed	https://docs.pytorch.org/docs/stable/distributed.html
PyTorch FSDP	https://docs.pytorch.org/docs/stable/fsdp.html
Google Cloud GPU overview	https://docs.cloud.google.com/compute/docs/gpus/overview
NVIDIA Triton	https://docs.nvidia.com/deeplearning/triton-inference-server/index.html
DeepSpeed ZeRO	https://www.deepspeed.ai/tutorials/zero/
ONNX Runtime	https://onnxruntime.ai/docs/
ONNX Runtime quantization	https://onnxruntime.ai/docs/performance/model-optimizations/quantization.html
MLflow tracking	https://mlflow.org/docs/latest/ml/tracking/
Apache Kafka	https://kafka.apache.org/documentation/
Apache Airflow	https://airflow.apache.org/docs/apache-airflow/stable/
Feast	https://docs.feast.dev/
Kubeflow	https://www.kubeflow.org/docs/
Prometheus	https://prometheus.io/docs/introduction/overview/
Grafana dashboards	https://grafana.com/docs/grafana/latest/visualizations/dashboards/
OpenTelemetry concepts	https://opentelemetry.io/docs/concepts/
Hugging Face PEFT	https://huggingface.co/docs/peft/index